FATİH ERDOĞAN

Software Engineer · Industrial & Logistics Systems · Applied AI

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PROFESSIONAL SUMMARY

Software engineer who turns complex real-world operations into production software. 4+ years across **industrial operations (MES/MOM, process and production data)**, **logistics**, and enterprise data, owning delivery from requirement discovery and architecture through **CI/CD** deployment on **Microsoft Azure**. Technical lead of an award-winning decision-support platform (**Brandon Hall Silver Medal**), builder of an executive-sponsored, **ML-backed procurement platform with human-in-the-loop workflows** running in production, and founding engineer of a **B2B logistics platform** with **agentic LLM features** (tool use, model routing, prompt caching, cost accounting). Core stack: **C#/.NET**, **React**, **TypeScript**, **PostgreSQL**, **Keycloak**, **Anthropic API**.

CORE SKILLS

- **Backend & Architecture:** C#, .NET 10 / .NET Core, Entity Framework Core, MediatR (CQRS), DDD, modular monolith, domain events, RESTful APIs, SignalR, RabbitMQ, Redis, workflow engines, Node.js, Express.js
- **Applied AI:** Anthropic Messages API, agentic tool-use loops, model routing, prompt caching, token and cost accounting, web-search grounding, LLM features in production workflows
- **Frontend & Mobile:** React.js (Vite), Next.js, TypeScript, JavaScript, React Native (Expo), three.js / WebGL, HTML, CSS
- **Testing:** xUnit, Testcontainers, NetArchTest (architecture tests), NSubstitute
- **Data:** PostgreSQL, SQL, MongoDB, Python, ETL/SSIS, SSAS, DAX, Power BI
- **Industrial & Enterprise Systems:** MES/MOM integration, PHD process-data systems, ERP (Microsoft Dynamics; SAP and ICRON data), production planning, process analysis
- **ML in Production:** ML model integration, confidence-aware human-in-the-loop UX, risk scoring on process data
- **Security & Identity:** Keycloak (OAuth 2.0, OpenID Connect, IAM, RBAC), ClamAV file scanning, Tesseract OCR
- **Cloud & DevOps:** Microsoft Azure (Container Apps, Static Web Apps, Blob Storage), Docker, GitHub Actions, CI/CD

EXPERIENCE

Senior OT MOM/MES Specialist — SOCAR Türkiye, İzmir

05/2023 – Present · Assistant Specialist (05/2023) → Specialist (06/2025) → Senior Specialist (07/2026)

- Led and built end to end (excluding the ML model itself) an executive-sponsored **procurement cost-optimization platform** that estimates target reference prices for materials and services; used by purchasing teams in tenders, bid evaluation, and supplier-discount negotiations, and delivering reliable results in production.
- Designed the **human-in-the-loop model-execution interface** around the ML model: surfaces meaningful outputs at each stage, asks users targeted questions when model confidence is low at critical milestones, and lets them intervene at any point.
- Built the **workflow engine** behind the full request lifecycle: manager routing, team assignment, revision cycles, and full or partial delegation of authority between managers and executors.
- **Technical lead** for the Decision Tool, a profitability and decision-support platform: defined the architecture and roadmap for real-time analysis and **risk scoring on production process data**; recognized with a **Silver Medal** at the Brandon Hall Group Excellence in Technology Awards.
- Developed **MES integrations** and data pipelines from process and production systems, enabling real-time production monitoring, operational transparency, and executive-level reporting.
- Turned operational requirements into scalable OT software solutions with cross-functional engineering, operations, and IT teams.

Founding Engineer — B2B Logistics Platform

03/2026 – Present · Sole engineer: web, iOS/Android and backend · ~2,700 commits · in QA ahead of go-live

- Built the platform single-handedly on the technical side, working with a product owner: a B2B logistics platform where shippers, carriers, and forwarders run RFQs, bidding, deals, and shared operations across road, sea, air, and rail; owned architecture, implementation, infrastructure, and delivery.
- Architected a **.NET 10 modular monolith** (DDD, Clean Architecture, **CQRS with MediatR**) of 13 domain modules, including shipment requests, bids, deals, tracking, dock scheduling, and documents; domain events keep modules consistent, and architecture tests plus ADRs enforce the boundaries.

- Designed an **agentic AI assistant** on the **Anthropic Messages API**: a server-side **tool-use loop** calling live road-routing tools, **model routing** (a small model for simple turns, a larger one for planning), tiered **prompt caching**, and **per-tenant token and cost accounting** that records counts, never tenant text.
- Shipped eight further **LLM features**: cross-language chat translation, message assessment and action extraction, offer explanations, dashboard and workflow drafting, and **web-search-grounded lane intelligence**.
- Multi-tenant security with **Keycloak** (OIDC, RBAC) and tenant isolation; **ClamAV** scanning and **OCR** on every uploaded document; real-time updates over **SignalR** with a Redis backplane.
- **3,000+ automated tests** (xUnit, Testcontainers PostgreSQL, NetArchTest); **React** (Vite) web app and **React Native (Expo)** mobile app; Docker and **GitHub Actions** CI/CD to **Azure Container Apps**.

Business Intelligence Analyst — NORM Digital, İzmir
10/2022 – 05/2023

- Built **SQL**-based reporting pipelines and **ETL** workflows from ERP systems feeding interactive **Power BI** dashboards for financial, operational, and sales metrics.
- Contributed to **AI-driven forecasting** solutions for strategic planning.
- Worked with the supply chain team on a **planning-optimization** project using ICRON and SAP data, modeling supply chain data for analytics.

ERP and Operations Specialist — YATSAN, İzmir
06/2022 – 10/2022

- Led ERP-based **production planning and inventory management** in Microsoft Dynamics.
- Contributed to process optimization in hands-on collaboration with manufacturing and planning teams; produced technical documentation and SolidWorks part models.

AWARDS

- **Silver Medal**, Brandon Hall Group Excellence in Technology Awards — Decision Tool (profitability and decision-support platform), SOCAR Türkiye; delivered as technical lead.

EDUCATION

Bachelor of Science, Industrial Engineering — Dokuz Eylül University, İzmir | 06/2022

LANGUAGES

Turkish — Native | English — B2 (Upper-Intermediate) | German — A2 (Elementary) | Japanese — A1 (Beginner)